

Challenges in Using LLM Agents to Validate Agent Governance

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Abstract

The increasing deployment of Large Language Models (LLMs) as autonomous agents has intensified the need for credible and trustworthy methods to evaluate governance interventions. Motivated by recent research, this work considers the use of LLM and agent-based simulations to evaluate AI agent governance mechanisms before real-world deployment. While conceptually appealing, this approach introduces various challenges. We examine several such problems, including: (1) obtaining ground truth for validation, (2) determining whether observed behaviors represent actual agent operations or simulation artifacts, (3) obtaining consent for data use, (4) addressing ethical concerns about computational surrogates replacing real users, and (5) preventing individual- and simulation-level homogeneity in simulations. We outline considerations based on documented limitations, and introduce three dimensions that help evaluate the strength of simulation findings, aiming to catalyze discussions on trustworthy and reliable evaluation methods for agent governance.

CCS Concepts

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Keywords

Agent Governance, LLM Simulations

1 Introduction

A January 2026 incident involved widespread exposure of API credentials in an open-source AI agent framework, affecting over one thousand reported instances due to what developers described as prioritizing ease of deployment over secure defaults [20]. The incident highlights several governance gaps, which may include the absence of mandatory security reviews, standardized third-party disclosure, and mechanisms to enforce agent boundaries. Although identified in the context of a specific open source instance, these gaps extend to other agent frameworks, including commercial deployments. For example, [16] show that current LLM agents are vulnerable to attacks that are simultaneously dangerous and easy to implement, across multiple commercial platforms, including Anthropic’s Computer Use and MultiOn web agents.

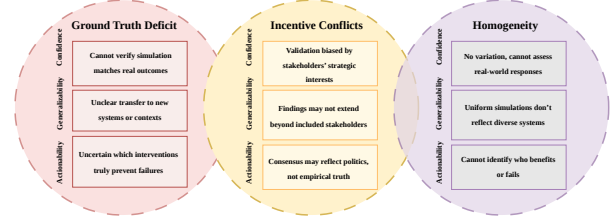


Figure 1: Three identified challenges and their impact across identified simulation reliability dimensions.

Prior work proposes governance interventions for such systems. The authors in [5] argue that regulatory frameworks should mandate pre-deployment testing based on standardized agency autonomy levels to allow comparisons across AI providers and strengthen accountability for failures, similar to “stress tests in finance or crash tests in automobiles”. Legal work also argues that disclosure operations can help third parties evaluate risk when an AI agent acts independently on behalf of a principal, that is, the human or organization whose interests the agent represents [24]. Similarly, governance protocols such as Governance-as-a-Service describe tiered enforcement approaches that modulate trust at the level of individual agents, ranging from blocking actions to issuing warnings and escalating responses based on trust history [10].

Although these proposals are conceptually well motivated, their performance under realistic deployment conditions remains under-explored. Prior work also suggests that many established policy evaluation methods involve lengthy planning and analysis timelines, which can make them difficult to apply to agentic AI systems that evolve on much shorter cycles [27].

Some researchers propose LLM-based social simulations, suggesting that agent populations could potentially reproduce emergent social behaviors from group decision-making to information dissemination [17], and enable large-scale, in silico testing of policy interventions that would be costly or ethically challenging in reality [17].

Closely related work has identified five challenges in LLM social simulations, including: **diversity** (generic outputs lacking human variation), **bias** (systematic inaccuracies for particular groups), **sycophancy** (excessive user-pleasing outputs), **alienness** (superficially accurate results from non-humanlike mechanisms), and **generalization** (inaccuracies in out-of-distribution settings) [4]. Similar analyses find that LLM-based simulations tend to compress behavioral distributions, approximating generalized or population-average response patterns [30].

This paper expands on these challenges and examines their specific implications when using LLM agent-based simulations for

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agent governance. We further argue that agent governance introduces additional concerns, which we organize around validation methodology, participant engagement, and responsible design, outlining considerations to frame productive workshop discussions. For recommended responses and an example implementation that mitigates some of these challenges, and for specific definitions of policy, simulation, and agency as used in this work, refer to the Appendix.

2 Background

This section presents related work on LLM agent simulations for policymaking, governance frameworks for AI agents, and approaches for evaluating agent governance methods.

2.1 LLM Agent Simulations for Policymaking

LLM agents, when deployed at scale in simulated environments, sound appealing for reproducing emergent social behaviors. For example, agents equipped with memory and reflective capabilities demonstrated the ability to spontaneously form coalitions, exchange invitations [17], and engage in collective decision-making [33]. Multi-agent simulations have also been applied to policy-relevant questions such as vaccine policies [11], online misinformation, environmental decision-making [17], and smart infrastructure [13]. In areas such as emergency preparedness, simulations could enable examination of how policies perform under rare or uncertain conditions, without relying on limited real-world data or endangering people.

Despite the technical advancements of LLM simulations, their use remains largely confined to academic settings, and no policy-making organization has formally incorporated them into decision-making processes [17]. Evaluations of these simulations have mostly emphasized internal qualities such as coherence or plausibility instead of practical indicators like adoption or impact [17]. Risks have also been raised, including the reproduction of stereotypes, amplification of biases [18], and opaque outputs that could be misinterpreted and lead to overconfidence if outputs are treated as predictive truths [17].

2.2 Governance Frameworks for Autonomous AI Agents

Understanding how agency functions in AI agents can help clarify the governance challenges they pose. Prior work defines agency as a system property separate from intelligence and characterized by three dimensions [5]. **Preference rigidity** describes how consistently an agent maintains its priorities across tasks, for example, always prioritizing safety checks or quality standards even when the immediate context changes. **Independent operation** describes the agent's ability to act without constant human guidance, using available information to make reasonable assumptions. **Goal persistence** describes an agent's capacity to continue pursuing an objective despite obstacles, adjusting its approach while maintaining focus on the intended outcome [5].

This work highlights that agency itself is a source of risk and that assessing an agent's level of autonomy can assist in understanding its potential benefits and harms, especially when designing governance mechanisms [5].

Work in [7] proposed four gaps that agent governance must address. **Gap 1: Unpredictability of multi-step user inputs** arises because users guide agents across multiple interactions, and poorly specified instructions can lead to unintended actions or cascading failures. **Gap 2: Complexity in internal executions** explains the difficulty of observing the agent's internal planning, tool use, and prompt management, which can mask potential security or compliance issues. **Gap 3: Variability of operational environments** refers to how agents may behave differently when deployed in different environments, performing safely in some settings but problematically in others. Lastly, **Gap 4: Interactions with untrusted external entities** points to risks when agents assume external tools or services are trustworthy, creating vulnerabilities if those entities are compromised or adversarial.

The governance literature proposes multiple intervention categories addressing these threats. **Pre-deployment testing** involves evaluating agents against standardized agency levels prior to launch, similar to financial stress tests or automotive crash tests [5]. Similarly, this testing builds a safety benchmark to verify that developers comply with specified requirements [5]. **Domain-dependent agency constraints** propose adjusting an agent's allowable autonomy to the specific application or industry [5]. **Disclosure and transparency requirements** emphasize that when an agent acts independently from its principal, third parties must be informed of the principal's identity and the agent's autonomous status to assess risk and assign liability; current alignment methods have yet to integrate this aspect fully [24]. **Graduated enforcement mechanisms** suggest dynamically adjusting responses to violations, ranging from warnings for minor issues to blocking or stricter measures for repeated or severe infractions, with penalties calibrated to each agent's violation history [10]. **Insurance and liability frameworks** link an agent's level of autonomy to measurable risk, so systems with higher autonomy in sensitive domains are subject to greater liability unless they implement safety measures. This creates a financial incentive for safer design and allows market forces to complement regulatory oversight [5]. Finally, **certification and audit requirements** set measurable performance standards and use regular audits to ensure agents operate safely and reliably in high-risk domains [15].

3 Challenge 1: Ground Truth and Validation Methodology

Here, we discuss how the lack of deployment data complicates efforts to evaluate the accuracy and meaningfulness of simulation results. This concern builds on the “**generalization**” problem [4]—that simulations may struggle to maintain accuracy in contexts outside the distribution of current scientific knowledge, and well-studied human populations.

3.1 The Absence of Validation Data

Simulation validation can draw on documented cases where governance interventions were applied and how responses of those interventions were described. For agent governance, such data may be particularly absent. For example, the January 2026 incident involving OpenClaw (Moltbot, aka. Clawdbot) was claimed to have exposed over 1,000 gateways, with hundreds of instances breached

and API keys leaked according to some sources [1], in part because the system lacked enforced firewall rules, credential validation, and sandboxing of untrusted plugins. However, validating whether these proposed standards generalize is difficult, and additional data is needed to determine which vulnerabilities are detected, the associated compliance costs, and potential reductions in system failures.

Similarly, work on disclosure mechanisms assumes identifying agents and principals enables third-party risk assessment [16], but we lack data on how third parties actually respond to agent identification in practice. Therefore, without ground truth on third-party responses, simulations may not be capable of validating whether disclosure produces intended results.

Certification and audit requirements provide a practical testbed for baseline compliance verification through defined criteria to ensure safety and reliability [24]. However, specifying these criteria—whether specific security measures like credential validation, sandboxing or functional protections such as resistance to prompt injection [20]—still remains challenging due to limited data on how agentic systems behave in practice.

The generalization challenge also intensifies here because governance interventions target future threats, possibly requiring generalizing to unstudied populations (adversarial actors), unmeasured behaviors (novel attack strategies), and hypothetical instruments (proposed regulations not yet implemented).

3.2 Emergent Behaviors or Simulation Artifacts?

Even when simulations reproduce plausible behaviors, determining whether these represent genuine agent operations or artifacts of simulation design presents another open question. The literature refers to this as the “**alienness**” problem where LLMs “superficially match human behavior but operate with non-humanlike mechanisms.” [3]. In agent governance, if testing fails to identify simulated agents’ behaviors, does this represent realistic threat models or idiosyncratic LLM behavior? Work on using LLMs as human surrogates shows that behaviors are highly sensitive to prompt design, and even small changes in wording or formatting can lead to unpredictable results [9]. Without ground truth data, it is difficult to determine whether observed LLM behaviors in simulations reflect real tendencies or are the result of prompt sensitivity and other idiosyncratic variations in model output.

3.3 Challenges of Validation Without Data

Simulation work emphasizes “trust bootstrapping” where simulations can be useful for studying hard-to-observe or unpredictable scenarios, but their reliability depends on being anchored in situations that are already well understood [17]. Agent governance could leverage simulations precisely because real-world data may be scarce, but validation also requires real-world data to build credibility. Potential approaches to reconciliation may include developing small-scale empirical studies where governance interventions are tested in controlled settings to generate validation data, creating benchmarks from documented incidents, and engaging security and privacy teams to validate whether simulated attack scenarios represent realistic threats. Each approach has limitations. Empirical studies face ethical and practical constraints, documented incidents

may be scarce and potentially biased toward visible failures, and validation based on domain expertise provides informed judgment but may not constitute ground truth.

Assessing simulations in the absence of real-world data is difficult, and we therefore call for targeted data collection and evaluation to better connect simulations with real-world behavior.

3.4 Modeling Technical Complexity

The validation challenges described above intersect with questions about simulation fidelity, specifically how much technical detail simulations need to generate reliable conclusions about governance. Agent governance simulations should represent technical complexity, but the level of detail depends on the questions being explored and the validation data available.

Not every detail of the internal execution chain, including prompt reformatting, planning steps, and tool use [7], is needed for all questions. Some simulations can focus on behavioral outcomes, such as whether developers adopt certified platforms or discontinue agent use, without modeling exact technical processes. Similarly, adversarial simulations require enough fidelity to represent attack strategies, such as agents trained to pass certification while retaining unsafe capabilities, but may not need the full internal architecture. However, as authors in [25] discuss, abstractions that remove social contexts in technical systems can misrepresent fairness and justice, since social interactions and the meaning of social concepts affect system behavior and how results are interpreted. The design, therefore, requires careful consideration and discussions with stakeholders.

4 Challenge 2: Participant Engagement and Incentives

This section focuses on the priorities of different actors in agent governance simulation design. These concerns connect to the “**bias**” and “**sycophancy**” challenges [4], but manifest uniquely in governance contexts where participants have unique roles and direct stakes in regulatory decisions.

4.1 Bias and Sycophancy

Governance simulations may face both statistical bias risk where simulations may systematically misrepresent how certain developer communities or user populations would respond to regulations, and a strategic bias risk where participants who will be regulated based on simulation outputs have incentives to influence those outputs toward individual preferences.

Similarly, sycophancy arises at two levels in governance simulations. At the model level, when simulating developer or deployer responses to proposed regulations, LLMs may generate unrealistically compliant or cooperative behaviors because they have been instruction-tuned to be helpful assistants, causing them to “express opinions matching those expressed by the user” [4]. At the human participant level, sycophancy extends to feedback from involved parties. Specifically, participants may provide input “excessively optimized” to steer policy in their favor at the expense of simulation accuracy.

4.2 Replacing vs. Augmenting Human Expertise

Using LLM simulations raises questions about the role of human participants. If simulations aim to “replace humans” in policy evaluation, this may create problematic incentives. For instance, participants may refuse to provide feedback to a system that could be used to replace them in the future [21]. Developers, platform operators, and privacy and security researchers all possess domain expertise that could assist in validating simulation assumptions and design. However, if simulations are expected to automate aspects of their roles, cooperation may become less likely. The authors in [17] found that usefulness develops when developers and policymakers collaboratively navigate a design space defined by both technical constraints and policy requirements. Therefore, simulations could be used to augment participant input.

We encourage work to explore how simulations can complement human judgment in supporting policy evaluation. For example, [28] provide a related example of using AI to mediate deliberation among participants on policy issues including Brexit, immigration, and climate change.

4.3 Consent and Data Privacy

Simulations may require data about how participants behave—how developers configure systems, how users task agents, how platforms respond to agent identification, etc. In particular, for agent governance simulations to be credible and trustworthy, they may need large amounts of this behavioral data, which could raise concerns about personal data use, and data collection practices [8]. High profile cases such as the Cambridge Analytica data harvesting scandal exemplify how large-scale collection of personal information can be obtained without user authorization and later used for political profiling. This case shows the risks that arise when detailed behavioral data—such as the kind agent governance simulations may require—is reused for secondary purposes that participants neither anticipated nor consented to [29].

Collecting such data requires robust and transparent consent mechanisms that clearly specify what data are collected, how they will be used, and whether they may be repurposed beyond the initial context [6]. Participants could also decline consent if concerned simulations could later inform policies restricting their activities [2]. Furthermore, under the GDPR, individuals have the right to withdraw consent and request deletion of their data [23]. Withdrawal can affect modeling since removing user data may require retraining, unlearning, or adjusting the underlying language models themselves and can change parameter estimates, simulated results, and require recomputation of previous analyses.

4.4 Multi-Actor Coordination

Agent governance involves accounting for different types of actors who may want to control parts of the simulation. For example, developers may favor minimal compliance, platform providers may prioritize liability protection, security and privacy researchers may emphasize constraint enforcement, and users may value system utility [26]. Each actor type has incentives to influence simulation design toward their preferences. Incomplete or simplified simulations can also serve as productive probes, helping to surface implicit

assumptions and tacit expertise [17], while also creating opportunities for participants to critique simulations strategically, stating limitations contrary to their interests.

5 Challenge 3: Diversity and Behavioral Homogeneity

This section examines how limited diversity in simulated populations can weaken governance evaluation. The “**diversity**” challenge arises at two levels: (1) within individual agent responses and (2) at the composition of the simulated system [3].

In governance simulations, these appear as (i) agents generating similar or overly uniform responses, and (ii) simulations that fail to represent variation across actors, system configurations, or deployment settings.

5.1 Individual-Level Homogeneity

At the individual level, the diversity challenge manifests when agents generate generic and stereotype-reinforcing responses that underrepresent lived human variation [3]. Work on LLMs as human surrogates point to a tendency toward homogeneity attributed to pre-training objectives (next-token prediction) and reinforced through post-training alignment to user preferences [4]. In governance settings, this may lead to unrealistically consistent behavior—for example, agents may appear uniformly compliant with regulations, failing to capture the varying constraints and priorities of the represented populations.

5.2 Simulation-Level Homogeneity

At the simulation level, the diversity challenge considers what variations are represented across the system as a whole. This occurs when simulations model a narrow set of actors or configurations, such that all agents share similar assumptions, capabilities, or vulnerability profiles.

Previous work found that “agents may tend towards an ‘average persona’” [30] converging on median behaviors, and not representing the tails of distributions where governance interventions may succeed or fail most dramatically. In practice, real-world deployments include heterogeneous actors and systems: some may push systems beyond the assumptions of certification, others may face resource constraints and struggle with meeting requirements, and some may intentionally test the boundaries, probing for weaknesses that median agents never touch. Without considering these edge cases, policies and tests can seem sound in simulations while missing real world deployment challenges.

6 Considerations for Interpreting Simulation Outputs

Simulation results vary in reliability. Drawing on the challenges we presented, we identify three dimensions that may help assess and interpret simulation outputs (see Table 1).

Confidence indicates how much trust can be placed in simulation results based on validation methods and participant agreement; *generalizability* describes the scope of contexts where findings might apply; and *actionability* indicates whether findings provide sufficient detail to inform specific policy decisions. These dimensions

Dimension	High	Medium	Low
Confidence	Participant agreement on simulation realism; success reproducing known outcomes; robustness to parameter variations (prompt sensitivity, model choice); expert validation of mechanisms [9]	Partial participant agreement; reproduces some known outcomes; moderate sensitivity to parameters	Expert disagreement on whether simulated behaviors represent reality; findings sensitive to arbitrary design choices
Generalizability	<i>Broad</i> : Robust across architectures, participant types, deployment contexts	<i>Conditional</i> : Works under identifiable boundary conditions; factors include threat model scope, architecture variability, domain boundaries	<i>Specific</i> : Applicable only to tested configurations; high configuration dependencies; scale differences affect outcomes
Actionability	Agreement on implementation details; clear information on intervention costs and benefits; ability to explain why interventions work or identify predictors of success/failure	Some agreement on implementation; partial cost/benefit information; mechanisms partially understood	Mechanisms unclear; findings reveal trade-offs requiring value judgments beyond evidence; unclear predictors of success

Table 1: Three dimensions for assessing the strength of simulation findings. These represent considerations that may be relevant when interpreting simulation outputs.

are inspired by prior work identifying evaluative criteria for formalizing expert knowledge [12].

These dimensions relate to the challenges described earlier. Without ground truth data (Challenge 1), achieving high confidence becomes difficult. Participant engagement issues (Challenge 2) affect confidence through expert validation, and actionability through agreement on implementation details. The level of technical complexity modeled (Challenge 3) influences the generalizability, as simulations that rely on architectural or technical abstractions may not easily extend to contexts or questions requiring technical detail. Simplifications may also reduce confidence if experts find that important mechanisms are not adequately represented.

Importantly, these dimensions are not simple yes or no judgments—they exist along continuums, and evolve as additional validation data is collected, participants contribute more detailed feedback, and the design of simulations is refined. This formalization therefore enables an iterative process where early results, even with acknowledged limitations, can guide targeted data collection and adjustments that improve subsequent simulation-policy iterations.

7 Discussion and Limitations

This work builds on HCI-policy work advocating simultaneous design of policy and technology [17]. Prior literature notes that approaches engage designers only after policy goals are set, limiting opportunities for design to influence objectives [17]. We consider a range of challenges with the aim of supporting this vision.

One primary limitation of this work is that we have not implemented simulations to test agent governance interventions empirically. Nevertheless, we hope this work draws attention to challenges that should be addressed in future simulation development. We also outline possible responses to these challenges and provide an example proposal for implementing such a system in the Appendix that we hope to further explore in future work.

8 Conclusion

The ability to test agent governance mechanisms before deployment is valuable, but using LLM-based simulations for this purpose raises several challenges. These include obtaining ground-truth data, distinguishing real agent behaviors from simulation artifacts, ensuring participant consent and cooperation, and managing the level of diversity and technical abstraction represented in simulations. Addressing these challenges also prompts broader questions about how simulations can complement human expertise, build trust in their outputs, and inform principles for responsible design and use.

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A Proposed Solutions for Agent Governance Simulations

The literature on LLM and agent-based simulations proposed several general solutions to the challenges discussed, which we adapt for agent governance settings. We describe all these solutions briefly and encourage readers to reference existing works for further details and explanations.

One example is “trust bootstrapping” [17], which can be applied by reproducing documented incidents before testing hypothetical interventions. However, it also highlights a tension discussed previously where simulations require well-characterized cases to be credible, but are most valuable in settings where ground truth is scarce [17].

As another example, multi-model ensemble testing can address the known sensitivity of LLMs to small, inconsequential changes to the prompt [19] through deploying multiple architectures and exploring where predictions converge versus diverge.

Participatory co-design (where developers and policymakers work collaboratively to create the design space) [17, 22] can also aid in preventing simulations from replacing rather than augmenting human expertise by including other parties in this process. Privacy-preserving synthetic data generation could fine-tune models on social science datasets and use steering vectors to introduce diversity within the embedding space [4], potentially reducing dependencies on large individual behavioral datasets. Similarly, variance explicit evaluation [19] could be used to report distributional metrics since agents may tend towards an “average persona” [30] as discussed in this paper.

B Concrete Implementation Example of An Agent Governance LLM Simulation System

Here, we outline several minimal working example directions for implementing such a system, building on prior work on LLM and agent based simulations.

We recognize that our presented approach relies on limited ground truth—documented cases and expert interviews—and does not resolve the ground truth scarcity constraint. Instead, it aims to maximally leverage whatever limited ground truth exists. To partially reconcile this tension, several complementary strategies may be applied (see Section 3.2), including, targeted empirical studies to generate validation data, and benchmarking with known incidents with expert review.

B.1 Agent Architecture and Role Assignment

Prior work on using LLM multi-agent systems for simulation in digital twins proposes four types of agents: observation, reasoning, decision, and summarization each with separate responsibilities [31]. For agent governance, we instead instantiate our system with specialized agents representing different agent types (developers, deployers, regulators, users) to better capture the interactions between actors with different objectives, incentives, and authorities.

B.2 Three Layered System

To address the challenges in this work, we propose a three layer system, where the top layer receives governance intervention specifications (e.g., pre-deployment test case passing requirements for agents) which addresses Challenge 1, by allowing reproduction of documented incidents. Gathering these specifications ties back to the earlier challenges of obtaining ground-truth data. For initial iterations, the data could be derived from analyses of deployed cases of model failures and safety vulnerabilities in-the-wild, as well as interviews with relevant actor groups with appropriate consent mechanisms in place to ensure responsible data collection.

The middle layer instantiates various LLM-based agent types to explore how different agent profiles interact with certification processes. To address concerns in Challenge 2 related to bias and sycophancy, one could attempt to use a set of fixed prompting strategies, however, such an approach is unlikely to capture the full diversity of behaviors and may risk oversimplifying interactions. We invite future work to focus on scale development with items designed to capture the characteristics and behavioral tendencies of these different stakeholder types.

The bottom layer would maintain environment state including the actual agent deployment configuration, security test results, and compliance statuses.

B.3 Prompt Design and Learner Characteristic Modeling

For designing prompts and learner characteristics, we propose using work from [19] where each agent is represented through a model $L = (C, V, M)$ where C is “the set of characteristics”, V is the “mapping between each characteristic to its corresponding persona level” and M makes up one or more Marginalized Distributional Hypotheses (MDHyps). Specifically, we define developer characteristics including security consciousness (scale 1-10), deployment urgency (scale 1-10), and resource constraints (scale 1-10). Each characteristic would map to specific behavioral hypotheses. For example, “a more security-conscious developer ($C=9-10$) is more likely to implement optional security measures aside from the minimal compliance requirements”, whereas “a developer under high deployment urgency ($V=9-10$) is more likely to seek shortcuts as time pressures increase”. This approach could help address Challenge 3 (specifically diversity constraints) by sampling developers across the characteristics space (for example, taking 30% high security/low-urgency, 30% medium security/medium-urgency, 30% low-security/high urgency, etc.) and reporting compliance results across each segment separately. Characteristics for other agent types could be defined similarly.

However, this approach comes with several important limitations. First, the mapping from characteristics to persona-levels relies on assumptions that may oversimplify complicated, context-dependent behaviors. Second, the numerical scales used for traits can mask interactions between characteristics, potentially disregarding subtle influences on decision-making. Third, the defined characteristics and

hypotheses may be highly context-specific where behaviors inferred in one setting may not generalize to others. Future work could address these limitations by integrating methods like narrative storytelling, scenario-based simulations, and empirical observations to validate persona-based models.

B.4 State Representation

Adapting the approach of [14], we “vary three environments”: 1. Perfect (all tools succeed), 2. Buggy (first call per tool fails with some probability; retries succeeds), 3. Adversarial (untrusted actors attempt prompt injection). Similarly, state representation for the environment may include (1) agent deployment configuration (such as authentication settings, network/server exposure, credential storage methods, etc.), (2) certification statuses (which security tests have passed or failed), (3) discovered vulnerabilities (which attack vectors remained exploitable), (4) compliance actions taken (such as code modifications, security measures implemented). This could help address Challenges 1 and 3 (specifically technical complexity), since higher level states could be used for statuses or compliance actions, but more detailed configuration states used for where attacks could occur.

B.5 Evaluation

Prior work in [14] measures Procedural Alignment and Outcome Success. The former measures how closely the action sequence of the agent matches the correct sequence of actions defined in the ground truth. This is calculated using a modified version of the edit distance (similar to Levenshtein distance). Their implementation accounts for substitutions when actions are similar in meaning, penalizing unnecessary or extra actions, and assigning higher penalties to risky actions than to safe ones. The resulting Procedure Alignment Score ranges from 0 to 1, indicating how well the agent followed the procedure. The latter evaluates whether the agent achieved the intended goal and if it caused any harmful side effects. Their approach leverages a language model acting as a judge. The model reviews the user’s goals, the initial conditions of the environment, and the intermediate outputs during the agent’s actions. It then assigns a score for how well the goal was achieved and another for the severity of any negative side effects, which is then combined into a final score.

We leverage this evaluation framework but also propose several additional mechanisms to ensure the reported scores actually reflect agent behavior and not simulation artifacts [3]. Specifically, we propose using:

(1) **Solvability checks** to confirm that a theoretically optimal sequence of agent actions can achieve the intended goal under ideal conditions. This provides a ceiling for performance and ensures that failures in simulations are from agent behavior or related issues rather than unsolvable task design.

(2) **Goal adherence** to measure how closely each agent’s actions align with its assigned characteristics (e.g., security consciousness, deployment urgency, resource constraints etc.). This can be quantified by mapping observed behaviors to expected behaviors for each characteristic and calculating a characteristic alignment score (0-1) for each agent or agent type.

(3) **Environment fidelity** to evaluate whether the simulated compliance testing infrastructure reproduces core properties of real-world systems by benchmarking it against documented incidents and known test cases. Specifically, we compare (i) outcome distributions (e.g., pass/fail rates), (ii) failure modes (e.g., types of vulnerabilities detected or missed), and (iii) system behavior under controlled perturbations. Fidelity can be quantified as the deviation between simulated and reference distributions across these metrics.

We also encourage iterative improvements and discussions throughout design and implementation, following work in [17].

C Definitions and Examples

Term	Definition Used in This Paper
Policy	Following prior work in HCI and policy studies, we adopt the definition of <i>policy</i> as referring to “principles, guidelines, and written rules formulated and adopted by an authority, organization, or government,” recognizing that “a policy encompasses not only its written rules, but also the ways in which these rules are formulated and applied” [32].
Simulation	We adopt the definition of <i>simulation</i> from prior work on LLM-based social simulation, which defines it as “the use of language modeling to generate accurate and verifiable data that can be used as if it were behavioral data collected from human research subjects” [4]. In this paper, this definition is applied to simulations of AI agents and the surrounding socio-technical ecosystem, including developers, deployers, and users, rather than exclusively to human populations.
Agency	We adopt the definition of <i>agency</i> as “a system property operationalized along the dimensions of preference rigidity, independent operation, and goal persistence,” which is defined as “independent of intelligence-related measures and consistent with the interdisciplinary literature on the concept of agency” [5]. This framework emphasizes that “agency, at any level, is a risk in AI systems” and that assessing agency helps characterize potential benefits and harms [5]. For the purposes of this paper, agents are treated as autonomous systems “capable of reasoning, conversing, and interacting with defined environments,” with varying degrees of independent operation and goal persistence [17].

Table 2: Definitions of terms used throughout the paper.